AGENTIC MUSIC TUTOR

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ABSTRACT

We propose an agentic system that helps self-taught learners master difficult pieces of music. The system supports learning in three ways: it simplifies difficult pieces, gives feedback on the learner's playing, and remembers the learner's past mistakes using LLM memory. An LLM orchestrator coordinates specialized tools: a transcription tool, an instrument player, a simplification agent that produces a ladder of progressively easier arrangements, and an assessment agent that listens to the learner, localizes mistakes, records them in memory, and provides personalized feedback. We will first evaluate each capability in isolation against single-model baselines to measure how much the agentic harness improves performance, and then demonstrate the full loop by teaching a person to play a difficult song.

1 PROBLEM DEFINITION AND MOTIVATION

Learning an instrument without a teacher is hard for three reasons. First, much of the music people want to learn exists only as audio recordings, without the notation needed to play it. Second, even when accurate notation exists, it is often far beyond the learner's level, and it may be available only for certain instruments; learners of less common instruments face an even harder task. Third, self-learners receive no feedback. They may not notice a missed note or a drifting tempo, and they can practice mistakes until those mistakes become ingrained. Commercial apps provide feedback, but only for a fixed catalog of songs. Human teachers address all three problems, but they are expensive, and many self-learners cannot commit to scheduled lessons.

Why an agentic system? An agentic system can reason over the outputs of its tools, make decisions based on the learner's past mistakes, and refine its behavior iteratively. Solving this problem end to end requires four capabilities that no single model provides. The first is *precise perception*: note-level transcription of complex audio, where we expect dedicated transcription models to be far more reliable than general audio-language models. The second is *generation and verification*: rendering notes back to audio and checking the result in a feedback loop, which a tool-calling harness supports naturally. The third is *symbolic reasoning over the score*, which requires structured, symbolic representations of the music rather than raw audio. The fourth is *adaptive multi-session planning*, which requires persistent memory so that the learner's past mistakes are not carried forward into later sessions.

The task is also naturally closed-loop. As the learner progresses to harder pieces, the agent must adapt the difficulty of its exercises and give increasingly precise feedback to keep the learner improving. This setting calls for an orchestrator that calls multiple tools and makes iterative decisions, which is exactly where we expect an agent to outperform a single model with a single prompt.

2 COMPARISON WITH EXISTING WORK

Existing tools and models already solve individual, one-shot music tasks. Commercial apps such as Suno (Suno, Inc., 2026) generate complete songs from a text prompt. Google DeepMind's open-source Magenta project (Google DeepMind, 2026) provides models for music generation and transcription, including the Onsets and Frames piano transcription model (Hawthorne et al., 2018),

which is trained on the MAESTRO dataset of aligned piano audio and MIDI (Hawthorne et al., 2019). Each of these tools, however, handles a single task such as generating music or extracting notes from a recording. No cohesive agentic system combines them to teach a learner. Other papers which are relevant to our work are Bittner et al. (2022); Dixon & Widmer (2005); Gardner et al. (2022); Kim et al. (2022); Müller (2015); Raffel et al. (2014); Ramoneda et al. (2022); Schick et al. (2023); Terao et al. (2022); Weck et al. (2024); Yao et al. (2023); Yu et al. (2023); Yuan et al. (2024).

3 Next Steps and Direction

Approach. We begin with piano, for which large datasets with ground-truth note annotations are available. An LLM orchestrator drives the feedback loop through the following tools and agents. A *transcription tool* converts audio to notes; we will validate it against ground truth, and at run time it can check itself by resynthesizing its output and comparing the result with the input audio. An *instrument player* renders notes as audio, on piano and on other instruments such as saxophone and guitar. A *simplification agent* produces an ordered ladder of easier arrangements of a difficult piece, with a rationale for each step. An *assessment agent* transcribes the learner's attempts, identifies wrong or out-of-place notes, and gives the learner feedback. Finally, a *mistake log* serves as persistent memory: it records the learner's mistakes, and the agents consult it whenever the learner starts new material.

Evaluation. We will evaluate the system in four ways. (1) *Transcription:* we will measure note-level F1 on a subset of MAESTRO, comparing the raw transcription model, the agent with self-checking, and a one-shot audio-LLM baseline. (2) *Perturbation detection*, our core evaluation: we will inject controlled errors of varying severity into ground-truth MIDI and test whether the agent detects them, measuring error-detection precision, recall, and localization accuracy. (3) *Simplification:* in a human evaluation, we will listen to the agent's arrangements and judge whether they are genuinely simpler, and we will also count the notes in each arrangement and assess how easy it is to play. (4) *End to end:* in a small human pilot, we will teach a volunteer a difficult piece they do not already know, identify failures, and use them to improve the system iteratively.

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A AI-USE DISCLOSURE STATEMENT

The project idea is the authors' own. The authors wrote the text of this proposal in their own words, and we used Claude Code (Anthropic) to improve the language, grammar, and formatting of that text, which included converting lists into prose and expanding the descriptions of the cited tools in Section 2. Claude Code was also used to help finalize the authors' ideas into concrete design choices (such as specific datasets, evaluation metrics, and baselines), to find and check the bibliographic details of

the references, and to set up the LaTeX template and project repository. The authors reviewed all of the text and take full responsibility for its content. No experiments have been run yet; we will disclose any AI assistance with code and experiments in later milestones.